ENGINEERING DRAWING EDR 101

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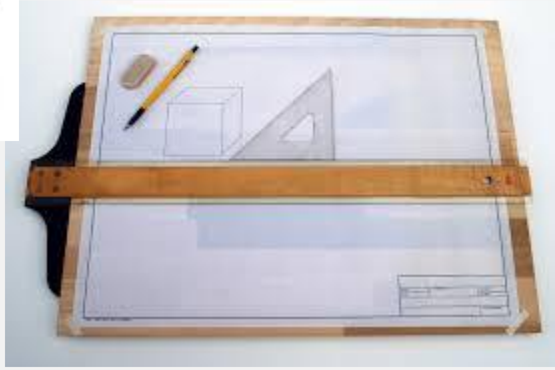
COURSE CONTENT

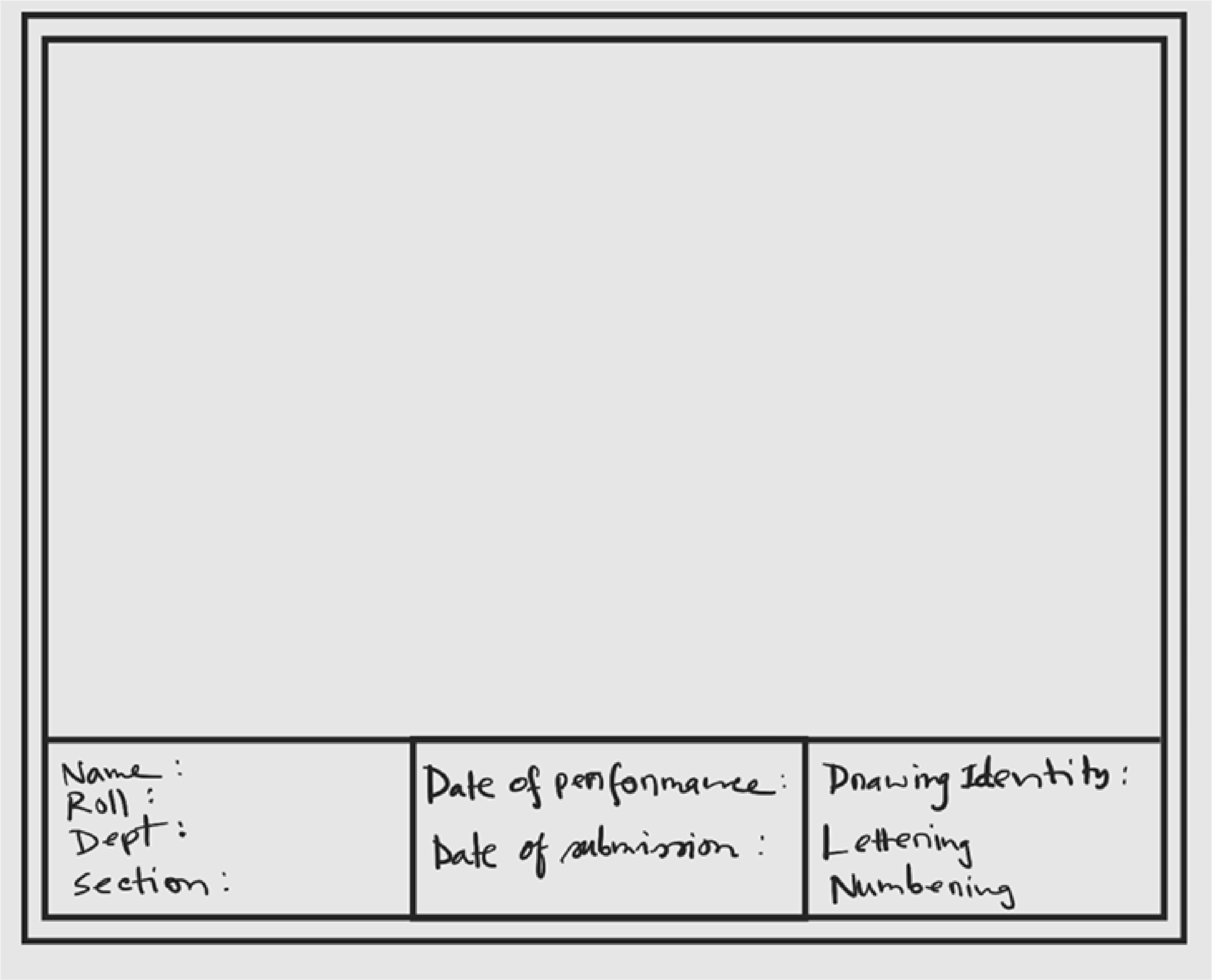
Lecture	Topics
01	Introduction to Engineering Drawing
02	Lettering, Numbering, Dimensioning, Drawing Scales
03	Plane Geometry: Pentagon, Hexagon, Octagon, Ellipse, Parabola, Hyperbola
04	Solid Object: Polyhedra, Prism, Pyramid, Cone, Cylinder, Sphere, Frustum, Truncated
05	Projection: Orthographic, Isometric, Oblique
06	Drawing isometric projections of solids; Drawing isometric views of objects
MID TERM – I	
07	Drawing orthographic projections of solids and different objects
08	Introduction to AutoCAD: Interface, Basic command, Layout, Creating different types of drawing, Dimension & Annotation, Printing & Publishing
09	Course wrap-up and evaluation
FINAL EXAMINATION	

DRAWING INSTRUMENTS AND MATERIALS

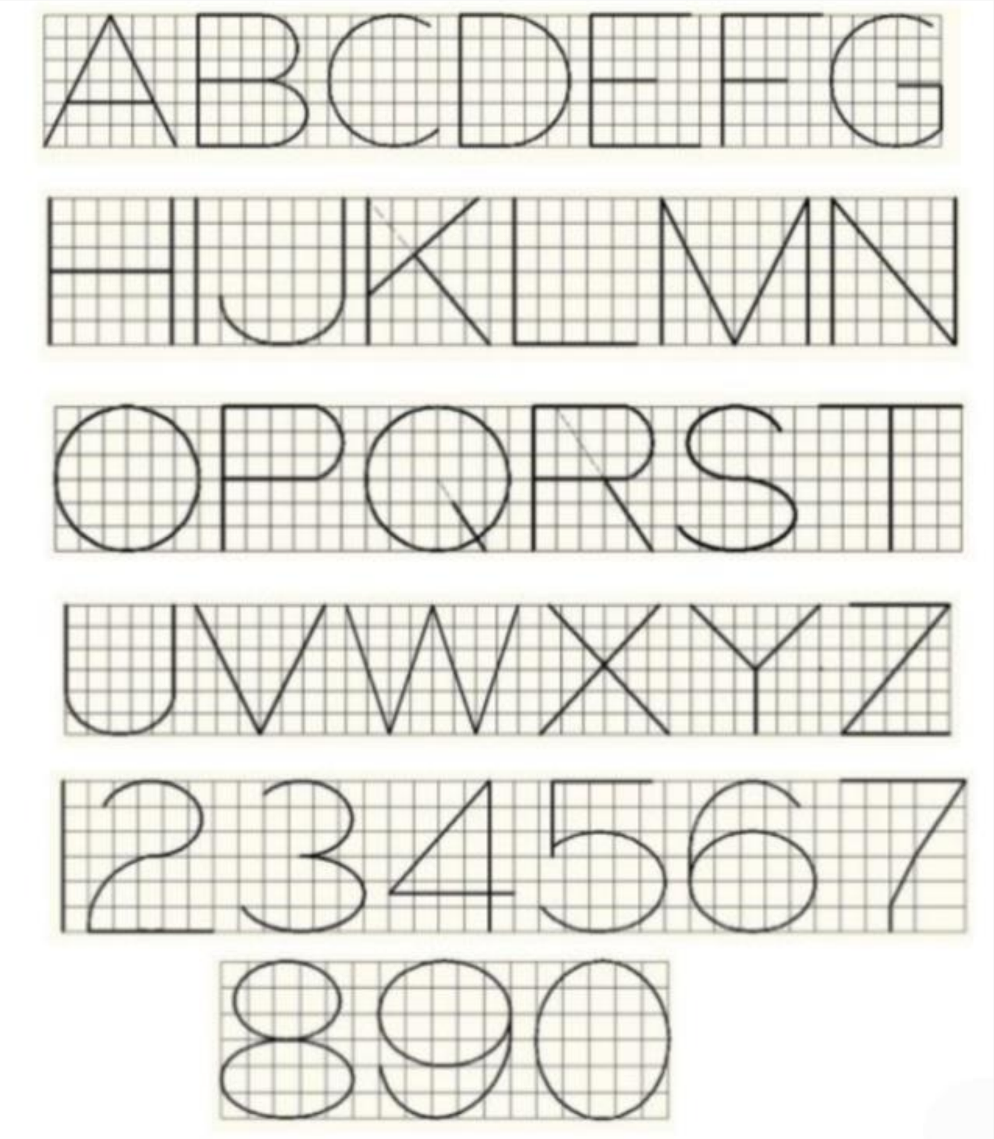
- Lead Pencils (2B and HB)
- Set squares
- Protractor
- Compass, Dividers, and Card board scale
- T square (i.e. “T” shape scale, 32" standard available in Bangladesh)
- Scale
- Drawing Paper (28" × 22")
- French Curve
- Scotch Tape

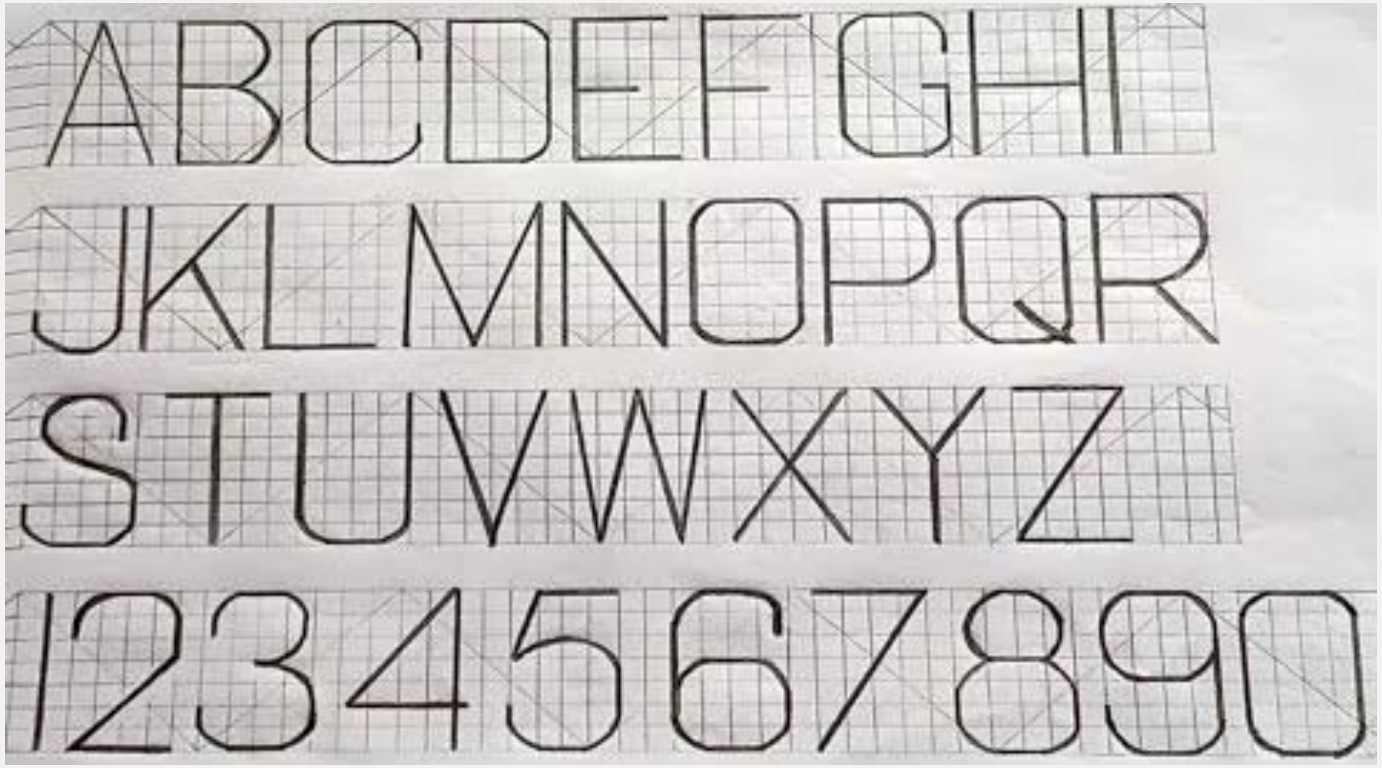


DRAWING LAYOUT



LETTERING AND NUMBERING





DIMENSIONING

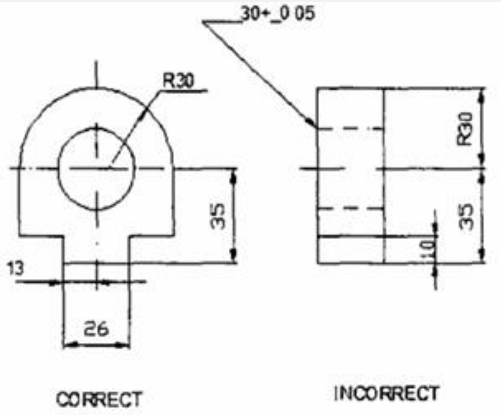


Fig. 2.14 Placing the Dimensions where the Shape is Best Shown

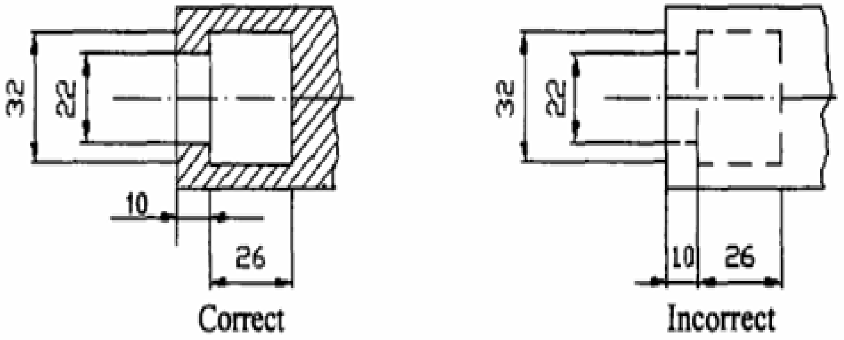


Fig. 2.16 Marking the dimensions from the visible outlines

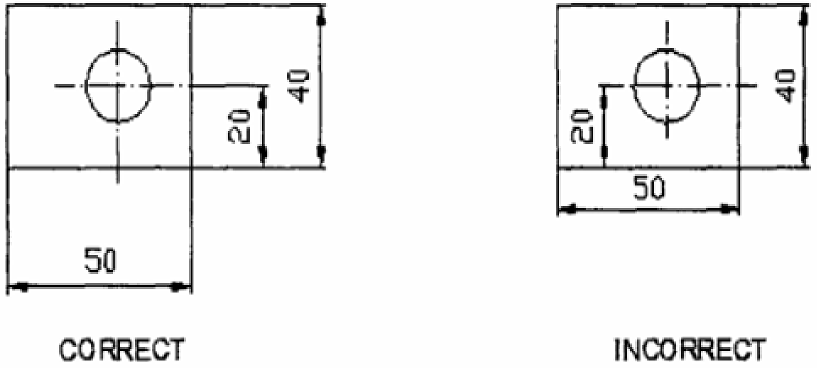


Fig. 2.15 Placing Dimensions Outside the View

DIMENSIONING

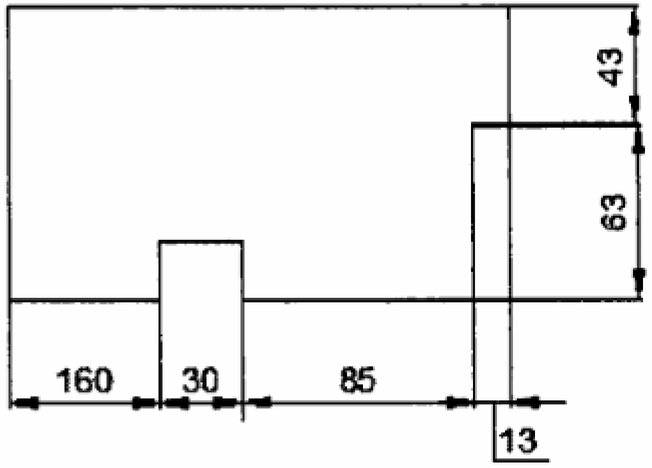


Fig. 2.33 Chain Dimensioning

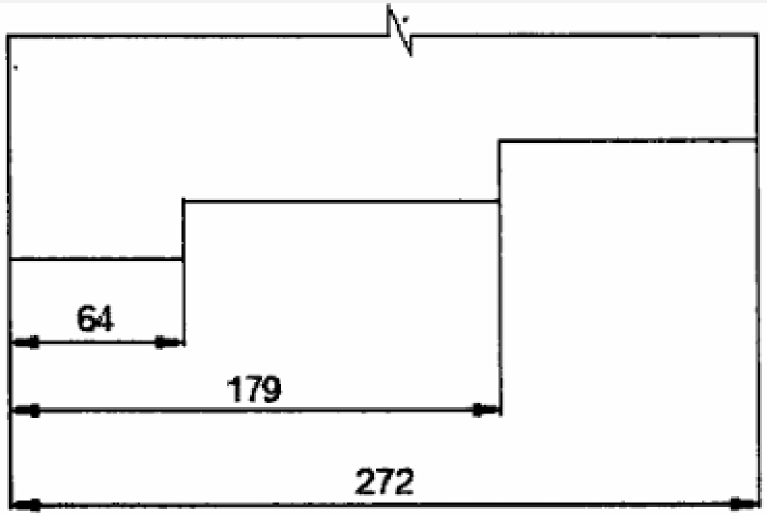


Fig. 2.34 Parallel Dimensioning

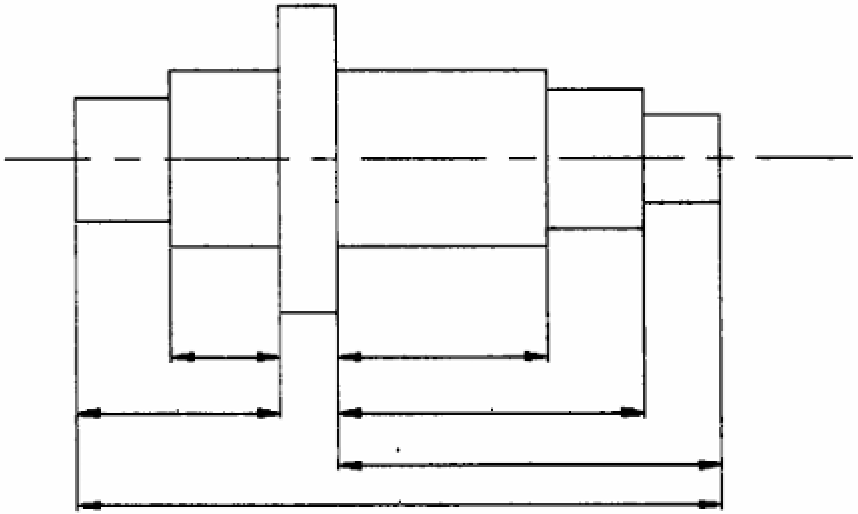


Fig. 2.35 Combined Dimensioning

DRAWING SCALES

- Scale 1: 1 for full size scale
- Scale 1: x for reducing scales (x = 10,20 ...... etc.)
- Scale x: 1 for enlarging scales.

- Representative Fraction: The ratio of the dimension of the object shown on the drawing to its actual size is called the Representative Fraction (RF).
- RF = $\frac{Drawing\ size\ of\ an\ object}{Its\ actual\ size}\ (in\ same\ units)$

THANK YOU